Emergency Maintenance Report — EMR-2025-004

CASCADIA POWER SERVICES INC.
Preventive and Reactive Field Services — Pacific Northwest Region
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Field	Value
Document ID	EMR-2025-004
Issued By	Cascadia Power Services Inc. (CPS)
Report Date	2025-10-15
Site	NW-PowerPool-WIND-003
Asset	WIN-10085 (Vestas wind turbine, 4.2 MW class)
Outage ID	OUT-2025-0476
Cascadia Work Order	WO-2025-10047
Authorization	Standard MSA (2023 multi-year), Schedule B — Reactive Response (4–8 hr)
CPS Crew Lead	K. Yamada, Lead Field Engineer
Distribution	Cascadia Operations · Cascadia Engineering · Cascadia AP

1. Scope of Emergency Work

CPS was dispatched 2025-10-07 06:14 PDT for a forced outage on WIN-10085. The trip was an overheating event at the generator coolant loop; nacelle ambient reached 71 °C against an alarm threshold of 60 °C. Because the event occurred during standard MSA hours and crew availability was confirmed, EPA-2024-11 was **not** invoked.

Authorized scope under standard MSA:
- Onsite isolation, cooling-loop inspection, oil and coolant sampling.
- Stabilization to return to service.
- Engineering recommendation for any follow-on planned work.

Outage duration to restoration: **226 hours**. Lost generation per Cascadia SCADA: 3,740.6 MWh.

2. Findings

Onsite arrival 2025-10-07 10:48 PDT (4h 34m from dispatch, within MSA Schedule B).

1. **Coolant pump P-2** showed degraded flow at 41% of rated. Internal inspection revealed impeller fouling consistent with deferred coolant change. Last full coolant change on file: 2024-02 (overdue under CPS-preferred 12-month cadence).
2. **Heat exchanger HX-1** showed 32% fin blockage from particulate, consistent with seasonal accumulation that should have been cleared in the spring PM cycle.
3. **Cooling-loop oil spectrometer** — CPS performed quarterly oil spectrometer testing per CSM-7. Wear metals within tolerance; coolant chemistry showed elevated copper (37 ppm) and reduced inhibitor concentration. CPS recommends Cascadia engineering confirm whether the quarterly CSM-7 spectrometer cadence remains the controlling procedure, given conflicting OEM guidance cited in EMR-2025-002.
4. **Generator bearings**, by contrast, are within wear tolerance. Vibration spectrum is clean; no spalling or micropitting indicators. The generator does not require replacement or refurbishment; it requires the cooling system to operate correctly.
5. The condition pattern is consistent with deferred cooling-system maintenance, not with end-of-life mechanical degradation. CPS does not see grounds for component replacement of the generator, gearbox, or main bearings.

3. Repairs Performed

Restoration and partial corrective work:

Item	Detail	Hours / Qty
De-energization, lock-out/tag-out, nacelle entry	Lead + 1 technician	14 hrs
Coolant drain, flush (3 passes), refill	320 L coolant, OEM-spec	38 hrs
Heat exchanger HX-1 cleaning, inspection	Particulate removed, fins straightened	26 hrs
Coolant pump P-2 impeller replacement (CPS PN VS-PMP-082)	1 unit	18 hrs
	Onsite + lab	11 hrs

Item	Detail	Hours / Qty
Oil spectrometer and coolant chemistry analysis		
Generator and main bearing inspection (no defects)	Borescope + vibration	22 hrs
Stepped re-energization	2 attempts, both successful	8 hrs
Documentation, photo log, PM cadence reset	—	9 hrs
Total labor		**146 hours**

Materials: 320 L Vestas-spec coolant; 1 impeller assembly (VS-PMP-082); HX-1 fin straightening kit; consumables.

4. Recommendations

The unit is fully restored to rated service. CPS makes the following recommendations, in line with our standing preference for refurbishment over replacement where mechanical wear is not at end of life:

1. **Cooling system overhaul, planned scope** — full coolant-loop refurbishment including pump P-2 rebuild (already done under emergency), pump P-1 rebuild, HX-1 and HX-2 full cleaning, expansion-tank pressure-relief replacement, and coolant-chemistry monitoring upgrade. Estimated planned scope: **$185K per turbine**. This is consistent with Cascadia Standard MS-2025-03 §5.2, which establishes cooling-system overhaul as a scheduled (not emergency) intervention. CPS Proposal CPS-Q-2025-074 will follow under separate cover.
2. **No grounds for generator, gearbox, or bearing replacement** at WIN-10085 based on the current condition. Replacement would not be cost-justified by observed wear.
3. Re-establish the 12-month coolant-change cadence at NW-PowerPool-WIND-003. The deferred 2024 cycle is, in our reading, a contributor to the present event.
4. CPS notes that the quarterly oil-spectrometer step under CSM-7 has been the cadence we follow; if Cascadia engineering elects to align CSM-7 with the OEM annual cadence cited elsewhere, we recommend formal MOC documentation and a controlled procedure change.

5. Invoice Summary

Line	Rate	Qty	Extended
Labor — standard MSA rate	$185.00/hr	146 hrs	$27,010.00
Subtotal labor			$27,010.00
Parts — coolant, 320 L	$42/L	—	$13,440.00
Parts — pump impeller VS-PMP-082	$7,800 ea	1	$7,800.00
Parts — HX-1 fin kit, gaskets, consumables	itemized	—	$4,920.00
Standard MSA parts mark-up	8% on parts subtotal	—	$2,093.00
Lab analysis — oil spectrometer + coolant chemistry	flat	—	$1,650.00
Crane / lift services (subcontract)	pass-through + 8%	—	$12,400.00
Travel, per-diem (2 techs × 9 days)	per MSA tariff	—	$6,420.00
Mobilization (MSA Schedule B, no premium)	flat	1	$4,200.00
Engineering review and documentation	$185/hr	24	$4,440.00
Post-restoration test report	flat	—	$3,800.00
Contingency reserve (drawn 0%)	—	—	$0.00
Coolant-chemistry monitoring upgrade allowance	—	—	$19,827.00
Total Invoice			**$108,000.00**

Closing

WIN-10085 returned to full rated service 2025-10-16 at 18:48 PDT. CPS recommends the planned cooling-system overhaul be issued under CPS-Q-2025-074. We do not see grounds for replacement-class scope on this asset at this time.

Signed	Name	Date
Crew Lead	K. Yamada	2025-10-15
CPS Engineering Review	A. Pellegrini, PE	2025-10-15

Signed	Name	Date
CPS Account Manager	S. Whittaker	2025-10-15

Attachments: Coolant chemistry report · oil spectrometer trace · HX-1 before/after photographs (10) · pump teardown photographs (6) · CPS-Q-2025-074 (cooling-system overhaul proposal).
CPS references: Vestas Service Manual VSM-4.2-CL §3.7 · Cascadia Standard MS-2025-03 §5.2 · CSM-7 §4.4.
Cross-references: OIR-2025-001 · EMR-2025-002 · CAR-2025-004 · RAM-2025-003.